

Investigation into the impact of a gamified cybersecurity learning platform

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ABSTRACT

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Education in cybersecurity often places an emphasis on theoretical knowledge over practical application, which results in a skill gap often found in beginner learners who can recognise the concepts but lack confidence in applying them (DSIT, 2025). This dissertation investigated the design, development and educational value of VulnLab, a gamified platform in the web browser used for learning web vulnerabilities, safe coding practices and introductory cybersecurity concepts through the use of structured learning and interactive challenges.

The system was developed using HTML, CSS, PHP, JavaScript and MySQL and deployed on the local system using XAMPP. Key features of the developed system include user authentication, interactive challenges (such as SIEM dashboard and web exploitation scenarios), skill-based progression, an experience points system, daily streak tracking, achievements, a game-based leaderboard and course unlocking based on user skill level, which measures competence. Learning is organised into three categories: beginner, intermediate and expert.

The project's motivation was the existing limitations in existing cybersecurity learning platforms for beginners, many of which include unstructured content, are too technically demanding for users to start learning or can be used as resources for hackers to exploit real systems. VulnLab addresses the limitations by providing a lightweight, browser-based alternative with guided progression and gamification grounded in Self-Determination Theory (Ryan and Deci, 2000) and Cognitive Load Theory (Sweller, 1988).

Evaluation done through the use of technical testing and user-gathered feedback concludes that a structured, gamified platform can support beginner cybersecurity learning by improving accessibility, maintaining engagement and supporting practical understanding through gamified challenges. However, the platform does not fully replicate a professional laboratory environment, but it demonstrates that a guided, lightweight approach can make a meaningful contribution to education in cybersecurity.

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1. INTRODUCTION

The growing number of cyber-attacks and data breaches has made cybersecurity one of the most important topics in modern computing, with demand for skilled professionals outpacing the supply (DSIT, 2025). As organisations have become more reliant on systems and online services, the demand for expertise in cybersecurity has increased significantly. However, an issue within the cybersecurity education is that many learners develop theoretical knowledge from universities and certification providers, but not enough practical practice, leading to

learners being unaware of how to put those skills into practice and use real tools (DSIT, 2025; Chothia and Novakovic, 2015).

This problem has become significant for beginners in the industry. For instance, a learner might understand concepts such as SQL injection, cross-site scripting, cookies, sessions, directory traversal or access control, but struggle to apply those ideas in a real simulated environment. Practical learning in cybersecurity is often more difficult to deliver than theoretical content because it can require significant infrastructure, an IT setup, specialised software and careful review of legal boundaries (Chothia and Novakovic, 2015). While tools exist that attempt to teach practical knowledge, they have significant faults for beginners, as explored in Chapter 2.

This dissertation investigates whether a browser-based learning platform in cybersecurity can offer a more accessible and structured approach to cybersecurity education (Chothia and Novakovic, 2015; TryHackMe, 2024; HackTheBox, 2024). This project is focused on the development of VulnLab, a web-based educational platform for cybersecurity that teaches web vulnerabilities, safe coding practices, and introductory cyber concepts through the use of structured gamified challenges. To keep users retained, the developed solution makes use of a reward system by using skill levels, daily streaks, achievements, gems and course unlocking through progression on the skill level scale.

While the original proposed solution focused strongly on web vulnerabilities, the final solution evolved into a wider cybersecurity learning platform with various topics in cybersecurity.

The change in scope reflects an important outcome in the development process. During the development of the website, it became clear that beginners might not benefit from starting straight away with advanced vulnerability topics alone. Instead, they might require a broader introduction to the topic before moving into more advanced areas. The final version of VulnLab, therefore, combines both foundational cybersecurity theory and practical learning.

The purpose of this dissertation is therefore in two ways. First, it documents the development and design of VulnLab in order to become a fully functional solution. Second, it discusses whether such a solution may contribute to the introductory education in cybersecurity by making concepts more structured, engaging and approachable.

The research question guiding this dissertation is: To what extent can a browser-based gamified learning platform improve accessibility, engagement and practical understanding of cybersecurity concepts for beginner learners?

The dissertation has been organised into several chapters. After the introduction, the context of the investigation explains why the problem is worth exploring and looks at the topic in greater depth. The literature review then looks at several relevant ideas in cybersecurity education, practical learning, cognitive load and gamification. The following chapters will examine aims and objectives, analysis and specification, methodology, project design and project implementation. The dissertation will then present the testing and evaluation, conclusion, and, in the end, a critical review and reflection on the overall outcome of this project.

2. THE CONTEXT OF THE INVESTIGATION

The issue being addressed by this project is the gap in theoretical knowledge and practical skill (DSIT, 2025; Chothia and Novakovic, 2015). Many new students are introduced to the field through textbooks, lectures, videos or certifications. This creates a theory-heavy focus that might lead learners to feel unconfident once stepping into the field and having to use actual tools or exploits. The core issue here is the lack of a structured approach in introducing students to practical concepts – this has been created because of the lack of infrastructure and appropriate environments to practice.

A number of platforms offer practical exercises and gamified interaction (TryHackMe, 2024; HackTheBox, 2024). However, these come with several disadvantages, such as assuming prior knowledge, hackers being able to exploit it, a lack of a structured approach or adding a technical overhead before users can use it. In such environments, the learner might struggle to know how to set it up, not know where to start, or not understand how one concept connects to another.

This suggests that the issue is not simply the absence of platforms teaching cybersecurity. Instead, the problem is created by the lack of guidance on these platforms and the lack of accessibility. A learner might understand that practical experience is important, but do not know where to begin or how a topic connects to another. This results in learners losing confidence early and viewing cybersecurity as hard to get into (DSIT, 2025; Chothia and Novakovic, 2015).

The context given makes the problem suitable for further investigation through the use of a structured educational platform. The project is based on the idea that new learners might benefit from an organised system that introduces concepts in a more organised way, is more accessible and rewards consistent engagement while reducing the overwhelming feeling that the other platforms give. This research, therefore, explores the intersection between cybersecurity education, software development, and beginner-focused learning design.

Furthermore, the investigation also looks at web security because these are one of the most common targets and because they can be easily demonstrated in a browser environment (OWASP, 2021). Topics such as login logic, requests, cookies, sessions, SQL injection and cross-site scripting can all be introduced this way easily.

This makes web security a suitable topic for beginner learning because the concepts taught are relevant in today's security and visible. Unlike some more specialised areas in cybersecurity, browser-based learning can be demonstrated easily. Learners are already familiar with logins and URLs, allowing the platform to build on already existing knowledge, resulting in easier learning.

During the development of the project, legal and ethical considerations were properly examined and considered. Practical cybersecurity education must be handled carefully, especially where simulated environments are involved, as these could be exploited by real hackers and hacked into if they use real systems (Chothia and Novakovic, 2015; TryHackMe, 2024; HackTheBox, 2024). For this reason, this project's simulations were made as local JavaScript-based simulations that can be run in a web browser rather than an unrestricted attack environment. This makes the system more suitable for educational use and easier to justify in terms of legality and ethical reasons.

The final solution was shaped by this context. Instead of trying to recreate an unrestricted attack environment or full-scale lab, this project provides a browser-based platform that stays local, safely supports cybersecurity learning, and teaches useful cyber concepts.

3. LITERATURE REVIEW

The literature review looks at the main ideas that were used for the development of VulnLab and explains why a browser-based platform using a gamified approach may be suitable for beginners getting into cybersecurity. This literature review further examines the wider need for practical cybersecurity learning, the educational value of hands-on environments, the importance of structured learning for beginners in cyberlearning platforms, the role of gamification and progression tracking to maintain motivation and the relevance of web vulnerabilities and safe coding practices. Overall, the literature review aims to identify the gap in cybersecurity learning approaches and explain how the final developed version of VulnLab closes that gap through structured progression, practical interaction and gamification.

3.1 Cybersecurity skills and beginner learning

Cybersecurity education has become more important as digital systems have become central to modern organisations and our daily lives. At the same time, the need for practical skills in cybersecurity has grown alongside the need for theoretical understanding. Recent UK labour market research has estimated that there were around 143,000 individuals in the UK workforce, but at the same time, it found a significant skill gap in general organisations and cyber businesses. Around 49% of businesses reported a technical security gap, while 30% reported gaps in more advanced areas. In the cybersecurity sector, 28% of the businesses reported a lack of technical skills among their existing staff. These findings highlight that cybersecurity education remains an important issue today and support the view that more practical exposure is needed rather than theory alone (DSIT, 2025). For beginner learners, this issue often stands out. A learner will be able to identify common cybersecurity concepts in theory, but still struggle to apply them in practice in a real environment. This suggests that beginner learning platforms should not only provide content, but also offer a structured and gradual learning process for learners (DSIT, 2025; Chothia and Novakovic, 2015).

The skill gap is not only found in the UK. Furnell and Bishop (2020) have talked about how cybersecurity globally has struggled to keep pace with the speed of technological change, making note that curricula often fall behind the threat landscape and that graduates often lack the practical skills needed to succeed. This further strengthens the case for creating such a platform like VulnLab, which tries to close the theory-practice gap that exists by delivering interactive content rather than only theory.

The importance of practical skills within cybersecurity rather than just theoretical knowledge has also been recognised at the policy level. Newhouse et al. (2017) developed the NICE Cybersecurity Workforce Framework, which aims to categorise cybersecurity work into defined roles and knowledge, skill, and ability (KSA) requirements. This framework focuses especially on the ability to perform tasks rather than simply recalling theory and specifies that being able to perform tasks is the distinguishing mark for a competent cyber professional (Newhouse et al., 2017). This further strengthens the argument that educational platforms should aim to develop task-based practical tasks alongside theory, which is exactly what VulnLab does with its challenge-first design.

3.2 PRACTICAL LEARNING IN CYBER-SECURITY

In cybersecurity education, practical ability is harder to teach than theory alone. Beginners can often spot basic concepts such as SQL injection, authentication failures, cookies, sessions or access control when they appear in written materials, but still struggle to apply those concepts in real life. Research conducted into practical cybersecurity learning has shown the value of a hands-on approach. Chothia and Novakovic, for example, have looked at a capture-the-flag virtual machine for

cybersecurity education and supported the educational value of such interactive challenge environments (Chothia and Novakovic, 2015). This is especially relevant for the developed solution because VulnLab was designed with the idea that practical interaction can make a huge difference and make security concepts more understandable and engaging. However, this literature also talks about how practical learning might not suit everyone. Many hands-on environments are only useful when learners have some technical confidence or understanding of how these tools work together. For beginner learners, especially, a practice environment can feel overwhelming if it is not guided properly and assumes prior knowledge. This is one of the ways in which VulnLab differs from other broad challenge gamified platforms. Research by Chothia and Novakovic (2015) supports the educational value of CTF-styled challenges, but also says that there are practical barriers that come with using resource-heavy approaches (Chothia and Novakovic, 2015). Likewise, platforms such as TryHackMe and Hack the Box show that browser-based cybersecurity learning can be effective; however, they still rely on cloud virtual machines or VPN-style lab access (TryHackMe, 2024; HackTheBox, 2024). VulnLab responds to the same educational need through a lighter browser-based approach, using locally deployed JavaScript simulated challenges instead.

The idea of using competitive exercises has a longer history than platforms like HackTheBox and TryHackMe suggest. Conklin (2006) looked at the educational value of security competitions in universities and found that hands-on competitive exercises improved student engagement and technical understanding compared to traditional theory-based instructions. VulnLab makes use of this with the use of leaderboards and challenge rooms for a variety of skill levels, increasing competition among users.

Kolb's Experimental Learning Theory proposes that effective learning happens through several experiences, such as concrete experience, reflective observation, abstract conceptualisation and active experimentation (Kolb, 1984). This reflects back to cybersecurity platforms where the learner is given a concrete problem, observes what happens when multiple approaches are tried, understands why a certain technique fails or succeeds, and then applies the understanding in the next attempt. Vuln Lab's design supports this by presenting learners with a concrete problem, providing hints to encourage reflection and requiring active experimentation to get the answer.

3.3 COGNITIVE LOAD AND STRUCTURED PROGRESSION

However, practical learning alone is not enough if it is not introduced in a way that is suitable for beginners. A major issue with many websites that teach practical concepts in cybersecurity is that they can overwhelm new users. Sweller's work on Cognitive Load Theory argues that learning is made more difficult when too much cognitive processing capacity is used by a practical task rather than having to understand the topic itself (Sweller, 1988). When learners are trying to understand a new security concept, follow

technical instructions, combined with a complicated interface, their learning might not be as effective. Sweller has suggested that conventional problem solving may consume too much processing capacity that could be instead used for learning (Sweller, 1988). This is important for cybersecurity as beginners are often introduced to complicated tools and workflows while learning new, unfamiliar topics. For this reason, structured learning is especially important. Cognitive Load Theory helps explain why guided tasks, hints and staged progression can be especially useful for beginners, as they reduce the amount of unnecessary mental effort spent on trying to explore the interactive challenge (Sweller, 1988). This idea influenced the final developed project through the use of skill-level-based progression, hints and different challenge styles intended to make learning more manageable. This can be seen in how VulnLab uses course unlocking and guided progression, where the learners are not expected to approach expert-level content from the start. This is consistent with Sweller's argument that excessive cognitive load can reduce the capacity available for learning (Sweller, 1988). The structured support provided by VulnLab can help beginners focus on one task at a time rather than managing the interface and learning the task at the same time.

3.4 GAMIFICATION AND MOTIVATION

Dicheva et al. identified the use of gamification in education as an expanding area with potential benefits for motivation and engagement, while suggesting that good design matters more than superficial reward use (Dicheva et al., 2015). Ryan and Deci's self-determination theory is also relevant here as it explains the motivation behind the use of autonomy, competence and relatedness (Ryan and Deci, 2000).

The final implementation has expanded on the use of gamification significantly. Besides the use of XP and levels, further gamification elements have been implemented, such as streaks, gems, achievements, course completion tracking and skill-level-based access to harder content. These features are useful not only to make the website feel like a game but also to provide feedback and visible progress to the user. Beginners need clear indicators that they are improving to keep them motivated and to have a visual basis of knowing how much they have improved. A challenge platform that has none of these indicators can feel discouraging for the users and not interactive, leading to demotivation. The solution proposed has clear, visible milestones, which were designed to support sustained engagement over time. Gamification is only effective when it supports the learning process rather than just being distracting. The reward systems (achievements, gems and XP) implemented on their own are not enough if the challenges used to gather them are confusing or poorly made. In VulnLab, gamification was therefore used as a supporting feature of learning to keep it entertaining rather than the main feature itself. Its purpose was to reinforce

the learning process, consistency and as a way for users to see the progress they made; however, the main challenges remain the central part of the website and the learning process. This approach is supported by Self-Determination Theory, which argues that motivation is strengthened when learners experience competence and autonomy (Ryan and Deci, 2000). In the developed project, features such as XP, achievements, streaks and learners being able to see their progression (using them) are meant to support the competence factor by making improvement easier to recognise, while being able to choose courses and the gradual unlocking of them helps the users gain a sense of autonomy. At the same time, Dicheva et al. (2015) note that gamification research often has a lack of strong evaluation, which shows us that the features should not be used as an assumption that they would improve learning and be assessed in the final project evaluation. (Dicheva et al., 2015)

3.5. WEB VULNERABILITIES AND SECURE CODING

A major focus in the original project was being able to teach web vulnerabilities. This has stayed the same in the final developed project, especially in categories such as SQL injection, XSS, Session State Security, API Security and Burp Suite. The relevance of these topics is supported by OWASP Top 10 (2021), which identifies major risks in web applications, such as broken access control, injection and identification and authentication failures (OWASP, 2021). The platform has broadened its scope, as mentioned before, going from only teaching web vulnerabilities to teaching wider theoretical concepts in cybersecurity; however, teaching web vulnerabilities still forms one of its strongest foundations.

This makes sense within the context of this dissertation because teaching web vulnerabilities is easier to show and explain. They can easily be demonstrated in a browser environment and connected directly with defensive thinking concepts. This can be useful for beginners, who can see when a vulnerability happens and how it can be prevented.

The structure VulnLab uses to teach web vulnerabilities is grounded in established security literature. Stuttard and Pinto (2011) give a number of web application attack techniques, such as SQL injection, cross-site scripting, authentication bypass, session management weaknesses and access control failures in detail. It demonstrates that these web vulnerabilities are not isolated cases but weaknesses that arise from predictable patterns in coding web applications. The topics that VulnLab teaches – such as SQL injection, XSS, session state security, and finding robots.txt to discover hidden files – link directly to the web vulnerabilities described by Stuttard and Pinto (2011),

confirming that the platform's content is based on real-world attack patterns and not just taught superficially.

By using the web browser, the platform can also teach both insecure behaviour and safer coding practices. A beginner can see, for example, how a weakness appears, and then see how it could be prevented through better coding decisions. This is especially relevant because the OWASP Top 10 (2021) continues to identify issues such as broken access control, injection and authentication failures as major web application risks (OWASP, 2021). VulnLab follows this by teaching browser-friendly topics such as SQL injection, XSS, session state security and API security in a way that remains practical enough to demonstrate clearly while still aligning with recognised industry risk areas. This idea is also supported by OWASP secure coding guidance, which recommends protections such as CSRF tokens, secure session management, stronger authentication controls and safer password handling (OWASP, 2024a; OWASP, 2024b; OWASP, 2024c). These ideas are present in the final implementation of VulnLab.

Overall, the literature above suggests that beginner cybersecurity education should use practical interaction to improve engagement and consistency. Additionally, the literature review suggests that these should be introduced in a structured, accessible and motivating way. The existing research supports hands-on learning, cognitive support through guided progression and the use of gamification to maintain engagement (Chothia and Novakovic, 2015; Sweller, 1988; Dicheva et al., 2015; Ryan and Deci, 2000); however, beginners still struggle when platforms assume prior learning. The final solution was designed to solve this gap by using a combination of browser-based challenges, progress tracking and motivational features.

4. AIM AND OBJECTIVES

4.1 AIM

The aim of this dissertation is to investigate the educational value of a gamified browser-based learning platform for teaching web vulnerabilities, safe coding practices and beginner cybersecurity concepts through the design, development and evaluation of VulnLab. The secondary aim is to determine whether a lighter simulated environment can provide a safer and more accessible alternative to infrastructure-heavy cybersecurity learning environments.

4.2 OBJECTIVES

The objectives of this project are:

1. To identify the existing limitations of cybersecurity learning found in relevant literature and existing platforms.

2. To define the critical success factors (CSFs) needed for the platform to become an effective beginner learning platform.
3. To design and develop a gamified cybersecurity learning platform that has structured progression and interactive challenges for engagement based on the CSFs.
4. To evaluate the effectiveness of VulnLab through technical testing and gathering user feedback.
5. To recommend the best practices for beginner web security education based on the final findings.

5. ANALYSIS AND SPECIFICATION LEADING TO THE PROPOSED DESIGN

The design of the final solution was made to support the educational needs of beginners. Beginner learners would struggle if the platform provided challenges without a structured, clear progression, as this would increase the likelihood of users becoming overwhelmed. As a result, the platform includes course categories, visible progression and challenge unlocking based on skill level.

The browser model was chosen over using real hardware because it was more suitable given the project timescale, more secure, as it cannot trigger real attacks, and it would save time from researching legal constraints we should consider. Additionally, by using a modular web-based design, it was easier to manage user authentication, progression tracking, challenges and reward systems within one platform.

Based on the findings identified in the literature review, five critical success factors were identified for this project to be successful and be considered an effective learning tool for beginner learners:

1. Accessibility – The platform must be easy to use by beginners without prior technical setup knowledge, which addresses the barrier identified by Chothia and Novakovic (2015).
2. Structured progression – Content must be organised based on difficulty and should be unlocked gradually based on gained XP. This is consistent with the findings within Sweller's Cognitive Load Theory.

3. Gamified engagement – Motivational features that support learning must be present (XP, streaks, achievements) and support learning rather than distract from it, as recommended by Dicheva et al. (2015)
4. Practical interaction – The platform must include hands-on challenges and not just theoretical knowledge, supporting the skill gap identified by DSIT (2025)
5. Ethical safety – Challenges must run in a controlled environment that cannot be exposed to hackers or exploited.

5.1 FUNCTIONAL REQUIREMENTS

The platform needs to provide the following core features:

- User registration, login and logout
- Database storing user progress
- Flag submission and validation
- XP awarding and rank calculation
- Gem rewards with leaderboard tracking
- Challenge unlocking based on skill level
- Streak tracking based on the user's daily activity
- Achievement unlocking based on user activity
- Dashboard-based summaries on rank, progress, streaks and completed tasks
- Course replay functionality
- Administrator page for managing users, content and roles

These requirements are necessary because VulnLab was not only intended to act as an educational platform but also as a gamified learning system. This means that the website needed not only to provide educational content but also to track learner activity, reward progress and unlock challenges based on user progress.

5.2 NON-FUNCTIONAL REQUIREMENTS

The main non-functional requirements are:

- Accessible through a web browser
- Suitable for beginner learning
- Secure handling of sessions and user credentials
- Modular code to be able to support maintenance and extensions
- Safe and controlled gamified challenges running on simulated local environments rather than real systems
- Good responsiveness during normal use
- Consistency of layout across all pages

These requirements were necessary because the platform was developed to be a safe educational prototype and not a production-ready solution. The solution reflects this through secure session configuration, CSRF protection, password validation and hashing, rate limiting of login attempts, modular PHP page organisation and reusable layout structure.

5.3 PROPOSED DESIGN AND JUSTIFICATION

The proposed design uses a three-layer web architecture: a presentation layer built with HTML, CSS and JavaScript, an application layer logic handled by PHP and a data layer that uses MySQL for the database storage, which is hosted locally through the use of XAMPP. The choice of the tech stack was made because it is lightweight, does not use any cloud infrastructure and suits building database-driven applications.

Alternative approaches were considered; one of them was to use Laravel, but this was rejected due to the additional configuration it would require. Using PHP with XAMPP was the most practical choice given the timescale of the project. Node.js was also evaluated but was ultimately rejected as it offered no meaningful advantage over PHP, as this project would be deployed only locally.

VulnLab is built using a modular structure, where each area – authentication, dashboard, courses, challenges, gamification, tournaments and administration were separated into different components. A central layout file maintains the structure across all the pages, making the website easier to change later. A dedicated security module centralises all security functions across the website.

The interactive challenge environments were built using independent JavaScript scripts, each being isolated from the rest of the platform. This means that by changing one of the challenges, interactive environments will not affect the others. The database was made to track user progress, with tables tracking user accounts, XP, skill level, streaks, course competition and achievements. The full database structure will be provided in Appendix IV.

6. METHODOLOGY

This chapter looks at the approach taken in designing VulnLab. It covers the overall development strategy used, data collection techniques, and the methods used to analyse and present results.

6.1 DEVELOPMENT APPROACH

An iterative, agile approach was used to develop this project. Instead of designing and building the whole website upfront, an incremental approach was taken. Each step focused on several features – such as authentication, progression (XP), individual challenge types and the gamification layer – all of which were built, tested and refined before the next step was taken. This helped ensure that each step met the standards required before moving on. This approach was also used because educational platform projects benefit from flexibility, as early prototypes might require revision based on further research and feedback.

Alternative approaches were considered. One idea was to use a waterfall model, but rejected because it would have required all design decisions to be finalised before any development began, which did not suit this project because a more exploratory nature was needed. A prototype-driven approach was also considered but was rejected because it lacked the structured planning needed to meet the objectives in time. The iterative/agile approach was found most suitable because it balanced the flexibility needed with structure for a single-developer project. This aligns with established software engineering guidance: Sommerville (2016) says that incremental and iterative development is well-suited for projects where requirements are not fully understood yet and where the system must change in response to feedback received and discoveries made during the development, both of which match this project well. (Somerville, 2016)

6.2 DATA COLLECTION TECHNIQUES

Two data collection methods were used to collect data in this project. The first one was for the technical testing, in which the functional and non-functional requirements (in Chapter 5) were compared against the implemented platform by using testing (in Chapter 8). The second was for the user evaluation, where volunteers with no cybersecurity experience were asked to use the platform and complete a questionnaire to assess its usability, engagement, and perceived learning value. Both methods used are described in more detail in Chapter 8.

The user questionnaire was designed using a five-point Likert scale for quantitative questions, and for qualitative questions, open-ended questions were used. Each question was linked directly to the Critical Success Factors that were identified in Chapter 5, making sure the data collected will be used to verify that the success factors were met. Only participants' first names were collected to satisfy the data minimisation principle.

6.3 DATA ANALYSIS

In terms of data analysis, quantitative results from the user evaluation were analysed by calculating the average score for each question using the scores given by the participants. Those averages were then compared against the CSFs and the objectives to assess whether the intended goals were met. The qualitative questions were reviewed by looking at the suggestions and identifying whether or not there were any recurring observations given. The evaluation and results can be found in Chapter 8.

6.4 DEVELOPMENT TECHNOLOGY

VulnLab was developed with LAMP stack technologies and was run locally with XAMPP. The core technologies used are shown in Table 2. The LAMP technologies used were: PHP for the server logic, HTML and CSS for the presentation layer, JavaScript for the interactive challenges, and MySQL was used to store user data. XAMPP was used because it is lightweight and requires no cloud infrastructure, and is well-suited to be used with PHP and MySQL. All development was made using two machines, a PC and a laptop running Windows.

7. PROJECT IMPLEMENTATION

This chapter describes how VulnLab was developed and taken from a proposed design into a working solution. This chapter covers the implementation of the platform core components and the key decisions made during the development, and the approach taken when building the interactive challenges.

7.1 PLATFORM ARCHITECTURE

The platform was created using a three-layered architecture as described in Chapter 5. A presentation layer (that uses HTML, CSS and JavaScript), an application layer (PHP), and a data layer (MySQL run via XAMPP). The codebase was given a modular structure, with

separated documents for authentication, dashboard, courses, challenges, gamification, tournaments and administration. A central layout file (layout.php) is used to maintain a consistent look across, reducing code duplication and making the code easier to maintain. A dedicated security file (security.php) was created to maintain all security functions across the platform, including session validation, CSRF token generation, rate limiting of login attempts to prevent brute force attacks, input sanitisation and password hashing using bcrypt. By doing this, we ensure that all functions are going to be applied consistently across all pages rather than being handled ad hoc with individual scripts.

7.2 USER AUTHENTICATION AND SESSION MANAGEMENT

The user registration was created with server-side validation that enforces a minimum password length of eight characters, the presence of an upper-case letter, and a unique username and password. Passwords are stored as bcrypt hashes using PHP's password_hash() function. Login was implemented with generic error messaging to prevent attacks such as username enumeration, and a rate-limiting mechanism was applied to automatically lock accounts after five failed login attempts. Sessions are managed through PHP's native session handling. All pages have session checks at the top of each script, redirecting users who are not authenticated to the login page.

7.3 GAMIFICATION

The skill level feature was implemented by keeping track of users' accumulated XP. On each successful challenge, a fixed XP value is given to the user and checked server-side against defined XP thresholds: Beginner (0-99), Intermediate (100-299 XP) and Expert (300+ XP). Courses are locked and unlocked by querying the user's current skill level against the required level for each one of the courses. The replay system was created by storing the completion state per user for a certain course. The awarding logic checks its state before giving XP and gems, therefore preventing double-awarding.

Daily streaks were created by recording the date of each user's last activity and comparing it to the current date. If the user has been active on the previous day, the daily streak will be incremented. If not, it will reset. Achievements were added by using certain conditions that are checked server-side, following a user's relevant action. Gems are also awarded with XP and stored in a separate column in the users table, which enables the leaderboard to rank users based on their gem balance.

7.4 INTERACTIVE CHALLENGES

Each challenge was built into its own JavaScript file. This decision ensured that the challenges were isolated from each other and from other parts of the platform, meaning modifying one challenge would not affect the rest of the website or other challenges.

The challenges consisted of a simulated terminal for Nmap challenges, a SIEM log, a web hacking simulation for SQLi, a robots.txt discovery exercise and a Burp simulation. Each course includes a hint system that provides guides without revealing the answer straight away, which is consistent with the structured progression approach found in the Cognitive Load Theory (Sweller, 1988).

The flag submission is handled on the server side using PHP, each flag being associated with a stored flag within the database. When a user submits a flag, the flag given is compared against the stored value and, if correct, the submission system is triggered, and XP is awarded. Direct URL access to the submission was restricted by checking that the request was made from an authenticated session that had loaded the course page previously.

7.5 DASHBOARD AND ADMINISTRATOR PAGE

The dashboard was made to display a summary of the user's skill levels and progress made within the platform. It shows the current XP, skill level, gem balance, active streak and the progress made within a range of skills and courses. These values are queried from the database each time the page loads, ensuring the values remain accurate after each activity is completed. The administration page is access-controlled using a role field in the users table, with non-admin users being redirected from the page if the role criteria are not satisfied. The admin panel allows an administrator to manage their users, modify challenge content and the values awarded (XP and gems) for each challenge.

5.4 TESTING STRATEGY

The testing of VulnLab will be divided into two phases.

Phase 1 covers all technical and functional testing, which verifies if each feature is working properly against the defined requirements in Chapter 5. Phase 2 includes user evaluation testing, which assesses whether the project achieves its intended educational value by using beginner participants. Together, the two phases will produce evidence required for Objective 4 (to evaluate the effectiveness of VulnLab) and provide the recommendations needed for Objective 5.

8.2 Phase 1 — Technical and Functional Testing

8.2.2 Technical Testing Table (Table 3)

Test no.	Feature being tested	Type	Test Description	Expected Result	Actual Result	Result
1	User Registration – Valid Data	Functional	Register with a valid username and email address	Account created; user redirected to login page with success message; bcrypt-hashed record stored in users table.	Redirected correctly, hash present in DB	Pass
2	Registration – Weak Password Rejected	Non-functional	Attempt to register with a password shorter than 8 characters or missing an uppercase letter.	Registration rejected with descriptive error; no record inserted	Error shown	Pass
3	Registration – Duplicate Email	Functional	Attempt to register with an email address that already exists	Error displayed; no duplicate record created	As expected	Pass
4	Login – Valid Credentials	Functional	Log in with a correct username and password, and also verify session is created, and the user is redirected to the dashboard	Session created and session cookie set	Confirmed through the use of browser-built-in tools	Pass
5	Login – Invalid Credentials	Functional	Log in with a correct username but a wrong password, then with a wrong username	Same generic error message shown	Same error shown	Pass
6	Rate Limiting on Login	Non-functional	Submit 6 failed login attempts within a short window for the same username	After 5 failures, the account is locked; the remaining lock time is shown in the error message	Locked after 5th attempt; message showed correct remaining time	Pass
7	CSRF Token Validation	Non-functional	Submit a login and a flag-submission form without the CSRF token field present	Both requests were rejected with a clear CSRF error; no processing occurs	Requests rejected with 'Invalid request' message	Pass
8	Session Protection – Unauthenticated Access	Non-functional	Navigate to dashboard.php, course.php and admin.php directly while not logged in	All three pages go back to login.php	Redirected on all pages	Pass
9	Direct Flag URL Access	Functional	Attempt to access the challenge flag submission endpoint (challenge_submit.php)	Access should be denied	Access denied as expected	Pass

Test no.	Feature being tested	Type	Test Description	Expected Result	Actual Result	Result
			directly via URL without completing the challenge			
10	User Logout	Functional	Click logout while authenticated, then try to press the browser back button	Session ended on logout; it does not go back to the logged in user	As expected	Pass
11	Dashboard – Stats Display	Functional	Complete one challenge, then check the dashboard for correct XP and gem balance, streak and completed-course count.	All of them display the expected result	All values correct	Pass
12	XP Award on Challenge Completion	Functional	Submit the correct flag for a new challenge, and then check the XP given	XP incremented	Both XP and gems updated correctly	Pass
13	No Double XP on Replay	Functional	Restart a completed course via the replay function and redo the course to “farm” XP.	No XP or gems awarded	Replay mode prevented double-award	Pass
14	Skill Level Progression: Beginner to Intermediate	Functional	Accumulate 100 XP to rank up	Skill level updates to Intermediate and new courses unlock	Rank badge and course unlocks updated correctly	Pass
15	Skill Level Progression: Intermediate to Expert	Functional	Accumulate 300 XP to rank up	Skill level updates to Expert and new courses unlock	Rank badge and course unlocks updated correctly	Pass
16	Course Locking – Intermediate Courses for Beginners	Functional	Log in as a Beginner user and navigate to the SQL injection page with the URL	Access denied	Access was denied	Pass
17	Daily Streak Tracking	Functional	Complete on two consecutive days	Streak counter shows 2	Streak incremented	Pass
18	Achievement – Cold boot (First Challenge)	Functional	Complete the very first challenge on a fresh account	Achievement unlocked and shown on the bottom-right corner as achieved	Achievement unlocked and displayed	Pass
19	Achievement – 3-Day Streak	Functional	Maintain an active streak for three consecutive days	Achievement for 3-day streak unlocked	Achievement unlocked	Pass
20	Achievement – Course Completion	Functional	Complete all challenges within the Offensive Security	Achievement 'Getting Started' unlocked	Achievement unlocked	Pass
21	Gem	Functional	Register two test	Users ranked in descending	Ranking correct	Pass

Test no.	Feature being tested	Type	Test Description	Expected Result	Actual Result	Result
	Leaderboard		accounts with different gem levels	gem order	after gem changes	
22	Terminal Challenge – Nmap Help Command	Functional	Open the Intro to Nmap course and type “ nmap –help”	Help output returned	Help output returned	Pass
23	Terminal Challenge – File System Navigation	Functional	Run 'ls', 'cat notes.txt', 'whoami' and 'pwd' in the terminal	The expected output was returned	All outputs matched	Pass
24	Web Hacking Challenge – Robots.txt Discovery	Functional	Navigate to robots.txt	Simulated robots.txt exposed showing /admin and /confidential_backup.pdf	Content displayed correctly	Pass
25	Web Hacking Challenge – SQL Injection Login Bypass	Functional	Submit a SQL injection	Challenge accepts the injection; flag awarded	Flag awarded	Pass
26	SIEM Challenge – Brute Force Detection and Block	Functional	Review the simulated SIEM logs and the attacking IP	It should show the expected flag	Correct IP identified and flag returned	Pass
27	Expert Lab – Burp Repeater Simulation	Functional	Load the Burp Repeater simulation in the Advanced Web Operations course and change the request.	It should show the expected flag	Flag shown	Pass
28	Tournament – Access by Rank	Functional	Attempt to enter the intermediate tournament as a beginner	The user should not be able to enter	Access denied	Pass
29	Admin Page – Non-Admin Access Blocked	Functional	Log in as a standard user and try to open the admin page via the direct link.	Access denied	Access denied as expected	Pass
30	Admin Page – Content Management	Functional	Log in as an admin user and modify a challenge	Changes should be displayed	Changes shown as expected	Pass

All 31 tests passed. This proves that VulnLab has achieved all the functional requirements defined in Chapter 5.

8.3 Phase 2 — User Evaluation Testing

8.3.1 Participant selection and questionnaire

User evaluation was made to gather feedback on the educational value, usability and engagement of VulnLab, which addresses Objective 4. In this case, three volunteers took part; all of these had very limited experience in cybersecurity, which matches the intended audience of the website. Participants agreed to provide feedback and were given the context of what the website is trying to achieve. The only data gathered in this feedback form was the first name, ensuring data minimisation.

The participants were:

Participant	Background	Experience in cybersecurity
Konrad	Computing student	None
Octavian	Non-computing background	None
Reymond	Computing student	Minimal — aware of basic concepts only

8.3.3 Questionnaire

Question nr	Question	Type	Dimension	CSF / Objective
Q1	The platform was easy to navigate without any previous experience	1–5	Usability	CSF 1 Obj. 4
Q2	The progression of challenges felt structured well	1–5	Usability	CSF 2 Obj. 4
Q3	The hint system was helpful	1–5	Usability	CSF 2 Obj. 4
Q4	The gamification features (XP, streaks, achievements, gems) made the platform more engaging	1–5	Gamification	CSF 3 Obj. 3
Q5	I feel I have a better understanding of cybersecurity concepts after using the platform.	1–5	Learning	CSF 4 Obj. 4
Q6	The interactive challenges were engaging and enjoyable.	1–5	Gamification	CSF 4 Obj. 3

Question nr	Question	Type	Dimension	CSF / Objective
Q7	The platform felt safe to use — I was not worried about doing something harmful.	1–5	Overall	CSF 5 Obj. 3
Q8	I would use this platform again to continue learning cybersecurity.	1–5	Overall	CSF 3 Obj. 5
Q9	What was the most useful part of the platform?	Open-ended	Overall	Obj. 4 / 5
Q10	What could be improved or added to make the platform more useful for beginners?	Open-ended	Overall	Obj. 5
Q11	Was there anything confusing or frustrating? If so, describe.	Open-ended	Overall	Obj. 4

8.3.1 Results

Question nr	Question	Konrad	Octavian	Reymond	Average grade
1	Is the website easy to navigate?	5	5	4	4.67
2	Is the website's content well structured?	5	4	5	4.67
3	Is the hint system helpful?	4	5	4	4.33

Question nr	Question	Konrad	Octavian	Reymond	Average grade
4	Does the gamification keep the user engaged?	5	5	5	5
5	Did you improve your understanding of cybersecurity?	4	4	4	4
6	Are the challenges engaging and enjoyable?	5	5	4	4.67
7	Does the platform feel safe to use?	5	5	5	5
8	Would you use VulnLab again?	4	5	5	4.67

In general, all the feedback received was positive across all areas, with an average score of 4.63 out of 5.

The highest rated questions were Q4 (gamification engagement) and Q7 (safety), both of which received a score of 5. This shows that the platform's gamification features were well received by the users as motivational tools, and that all participants felt like the website was providing a safe environment to practice without the users being scared of attacking real systems. These results provide direct evidence that CSF 3 (the gamified engagement) and CSF 5 (ethical safety) were achieved.

Q1 and Q2, which look at the accessibility and structured progression, achieved an average of 4.67, supporting that the CSF 1 and CSF 2 were successfully met. Octavian, who had no computing background, rated the navigation 5 out of 5, suggesting that the platform will be easily accessible to beginner learners. Q5, which looked at perceived learning improvement, recorded the lowest rating with an average of 4, but this still stands as a strong result. It's also important to note that a single session on the platform cannot help users retain the knowledge; this result should therefore be looked at as encouraging.

8.3.1 Open-ended questions

The open-ended questions were created to provide qualitative detail to support the rated scores.

For Q9, all three participants said that the interactive challenges were the feature they found most valuable. Konrad said that being able to try exploits in such an interactive, simulated environment made abstract concepts easier to understand. Octavian, despite having no computing background, said that the structured course progression helped him understand how topics connect to each other and this helped him learn gradually. Reymond said that the hint system was particularly useful to him when he got lost, and mentioned that he was glad the answer was not given away, but he was guided more towards the correct answer.

For Q10 (suggestions for improvement), participants looked at how the platform can be expanded. Konrad suggested adding more network challenges could be particularly useful, as it would make the platform more comprehensive. Octavian has suggested that a short introduction could be useful for the new learners to learn how to use the platform. Reymond suggested that a progress summary at the end of each course could be useful, as it would reinforce learning.

For Q11 (something confusing or frustrating), the responses were positive. Konrad and Reymond found no issues. While Octavian noted that a couple of the challenge descriptions are not clear enough, and he found some of them confusing.

Summary

The user evaluation strongly suggests that VulnLab has achieved its intended educational value and engagement goals for the targeted audience. Every test averaged over 4, and the

open-ended questions confirmed that the platform is accessible, engaging and safe. All the suggestions gathered are good future improvements and don't represent fundamental flaws.

8. CONCLUSIONS

This chapter shows all the findings made in this dissertation and looks at each of the objectives in order to be able to conclude the research in question: To what extent can a browser-based gamified learning platform improve the accessibility, engagement and practical understanding of beginners in cybersecurity?

9.1 SUMMARY OF FINDINGS

The literature review was able to identify a well-documented gap between theoretical knowledge and practical skills in cybersecurity education, supported by evidence from DSIT (2025), Chothia and Novakovic (2015), and Furnell and Bishop (2020). Some platforms have tried to address the gap, such as TryHackMe and HackTheBox, but they introduce accessibility barriers for beginners through the need for technical setup and assumptions of prior knowledge. VulnLab was created to address this problem by delivering a browser-based system with guided, gamified challenges that require no technical setup.

The five critical factors previously identified in Chapter 5 – which are accessibility, structured progression, gamified engagement, practical interaction and ethical safety – were checked by the use of technical testing and user evaluation, which are present in Chapter 8. All tests passed with scores over four, indicating the learners found the platform accessible, engaging and safe. Q4 (gamification) and Q7 (safety) both have received a perfect average of 5, and Q1 (accessibility) and Q2 (structured progression) both scored an average of 4.67, directly showing that CSF 1,2,3 and 5 were achieved. Q5 (perceived learning improvement) has received the lowest result with an average of 4, but this still represents a strong result, given that the users have only used the platform one time.

9.2 CONCLUSIONS AGAINST OBJECTIVES

The objective number 1 (to identify the existing limitations in cyber security) was addressed in the literature review available in Chapter 3 and the analysis in Chapter 2, which looked at the inaccessibility of the existing platforms, the assumption that users have prior knowledge and the absence of structured learning. Objective 2 (to define critical success factors) was achieved in Chapter 5, where the CSFs were defined to guide the design and evaluation of the platform. The objective 3 (to design and develop the platform) was met through the analysis, specification and implementation

available in Chapters 5, 6, and 7, resulting in a fully functional solution with structured progression, interactive challenges and a gamification system. Objective 4 (evaluate the effectiveness of the platform) was identified in Chapter 8 through the use of technical testing and user evaluation that used three independent non-experienced participants. Objective 5 (to recommend best practices) will be addressed in the next section below (Section 9.3).

9.3 RECOMMENDATIONS FOR CYBERSECURITY EDUCATION

Based on all the findings, the following recommendations are made for the design of beginner cybersecurity learning platforms. First, accessibility should be a priority as beginners lack the technical knowledge or the infrastructure to run such tools. This represents a barrier in beginner cybersecurity learning that affects the audience they are trying to help. Second, structured learning is essential. By skill-locking certain courses, we can make sure the learner has enough knowledge to progress. By simply presenting all content available or allowing users to jump to certain courses that are more advanced, there is a risk that users might be overwhelmed, a risk which is grounded in Cognitive Load Theory (Sweller, 1988; Paas et al., 2003). Content should be organised by difficulty and unlocked after certain knowledge has been reached, as VulnLab does. Third, gamification should have a purpose rather than being applied superficially. Features such as XP, streaks and achievements can help when they provide feedback rather than just being used as a distraction (Hamari et al., 2014; Dicheva et al., 2015). Fourth, the hint system should be applied in an incremental way. Hints that narrow a problem down are more useful than those that answer away, as they allow the users to build their own understanding. Fifth, ethical safety must be built into the design. Simulated environments are most effective as they run locally and cannot connect to real systems, providing learners with a safe environment to learn.

9. CRITICAL REVIEW AND REFLECTION

This chapter looks at the research and development process critically, identifying both its strengths and its weaknesses in the conception, design, implementation and evaluation of VulnLab, and considers what questions the project has raised for further work.

10.1 STRENGTHS

One of the most significant strengths of this project was its clear linking with available research. By using the design decisions in Cognitive Load Theory (Sweller, 1988; Paas et al., 2003), Self-Determination Theory (Ryan and Deci, 2000) and gamification research (Dicheva et al., 2015, Hamari et al., 2014), the project progressed from a simple functional web application to a platform that can justify its design decisions against established evidence. The theoretical foundation also strengthened the evaluation, because the critical success factors and questionnaire were created based on the literature review, making it easy to map user feedback onto the theoretical claims from the literature review.

The second strength is how complete the platform is. Despite being a single developer project developed within a specified academic deadline, VulnLab delivered a working prototype that integrates user authentication, role-based access control, a full gamification system, interactive challenges, and an admin system. Additionally, all thirty tests were passed, demonstrating that VulnLab met its defined requirements. The security implementation is also strong for a project of this scale: bcrypt hashing, CSRF protection, rate-limiting login system, generic error messages without giving away certain vulnerabilities and session-based access. These successfully align with OWASP guidance (OWASP, 2024a, OWASP, 2024b, OWASP, 2024c).

10.2 WEAKNESSES

The biggest weakness of this project was the number of participants used for the user evaluation. Three participants are a small number to draw conclusions from, and all three people were known to the author, which introduces a familiarity bias. A stronger evaluation would involve a larger, more diverse group that is recruited randomly. The results are encouraging, but these cannot be treated as statistically representative. Dicheva et al. (2015) noted a similar weakness in most gamification research literature, noting that many platforms lack rigorous independent testing, and this project shares this limitation.

A second limitation is the authenticity of the challenges present in VulnLab. These are JavaScript simulations that do not expose the learners to the true complexity of these tools. The Burp Repeater, for example, gives the users an introductory lecture in HTTP request manipulation but does not let the users use the actual Burp Interface, which they would actually need to use in real life. This limitation between the simulated environment and the real tools is a trade-off of using a browser-based, ethically safe approach. Given this, VulnLab is rather described as an introductory supplement that could evolve into a real solution in cybersecurity rather than a replacement for more advanced tools for now.

A third limitation is the way the platform was tested. The platform was tested only in a local environment. No real penetration testing using tools was performed, nor was long-term usage monitoring performed. Additionally,

security measures that were implemented are good enough for a prototype, but a full production deployment would require further hardening before real users are able to use it.

10.3 FURTHER RESEARCH

This project has created several questions that would benefit from more investigation. For starters, the most imminent question is whether using VulnLab more often, for a longer period, rather than a single session, would produce any significant learning for the users and improve their confidence in cybersecurity. A second question for further research would be whether adding more network-focused challenges would make the platform more useful as a stepping stone to more advanced environments. Finally, concerns the potential transition from the locally-run XAMPP to a cloud environment. A publicly available instance of VulnLab would allow a much larger diverse group of users to access the platform and allow long-term use data to be collected, supporting the evaluation.

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